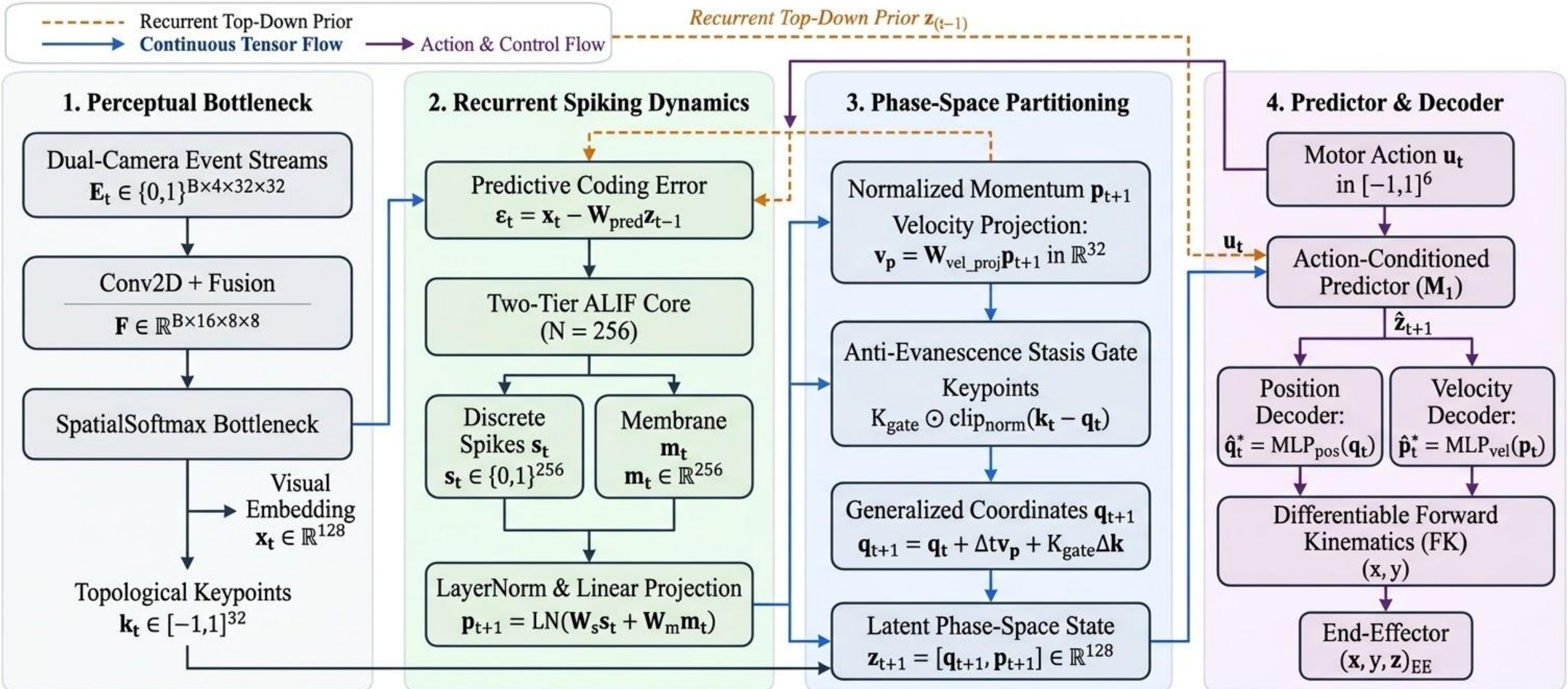

Recurrent Top-Down Prior
Continuous Tensor Flow
Action & Control Flow
Recurrent Top-Down Prior $\mathbf{z}_{(t-1)}$
1. Perceptual Bottleneck
Dual-Camera Event Streams
$\mathbf{E}_t \in \{0,1\}^{B\times4\times32\times32}$
Conv2D + Fusion
$\mathbf{F} \in \mathbb{R}^{B\times16\times8\times8}$
SpatialSoftmax Bottleneck
Visual Embedding
$\mathbf{x}_t \in \mathbb{R}^{128}$
Topological Keypoints
$\mathbf{k}_t \in [-1,1]^{32}$
2. Recurrent Spiking Dynamics
Predictive Coding Error
$\boldsymbol{\varepsilon}_t = \mathbf{x}_t - \mathbf{W}_{pred}\mathbf{z}_{t-1}$
Two-Tier ALIF Core
(N = 256)
Discrete Spikes $\mathbf{s}_t$
$\mathbf{s}_t \in \{0,1\}^{256}$
Membrane $\mathbf{m}_t$
$\mathbf{m}_t \in \mathbb{R}^{256}$
LayerNorm & Linear Projection
$\mathbf{p}_{t+1} = \mathrm{LN}(\mathbf{W}_s\mathbf{s}_t + \mathbf{W}_m\mathbf{m}_t)$
3. Phase-Space Partitioning
Normalized Momentum $\mathbf{p}_{t+1}$
Velocity Projection:
$\mathbf{v}_p = \mathbf{W}_{vel_proj}\mathbf{p}_{t+1}$ in $\mathbb{R}^{32}$
Anti-Evanescence Stasis Gate
Keypoints
$\mathbf{K}_{gate} \odot \mathrm{clip}_{norm}(\mathbf{k}_t - \mathbf{q}_t)$
Generalized Coordinates $\mathbf{q}_{t+1}$
$\mathbf{q}_{t+1} = \mathbf{q}_t + \Delta t\mathbf{v}_p + \mathbf{K}_{gate}\Delta\mathbf{k}$
Latent Phase-Space State
$\mathbf{z}_{t+1} = [\mathbf{q}_{t+1}, \mathbf{p}_{t+1}] \in \mathbb{R}^{128}$
4. Predictor & Decoder
Motor Action $\mathbf{u}_t$
in $[-1,1]^6$
$\mathbf{u}_t$
Action-Conditioned
Predictor ($\mathbf{M}_1$)
$\hat{\mathbf{z}}_{t+1}$
Position Decoder:
$\hat{\mathbf{q}}_t^* = \mathrm{MLP}_{pos}(\mathbf{q}_t)$
Velocity Decoder:
$\hat{\mathbf{p}}_t^* = \mathrm{MLP}_{vel}(\mathbf{p}_t)$
Differentiable Forward
Kinematics (FK)
(x, y)
End-Effector
$(\mathbf{x}, \mathbf{y}, \mathbf{z})_{EE}$